

ARTICLE

Encouraging crossover voting in the 2024 presidential primary

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Abstract

Will voters participate in the primary of a party they oppose to prevent the nomination of a candidate they fear? Partnering with a political action committee, we conduct a first-of-its-kind, large, preregistered field experiment ($N = 83,902$) in the lead-up to the 2024 Republican presidential primary in New Hampshire. A specialized get-out-the-vote intervention increases turnout in the Republican primary among undeclared voters who are modeled as likely to vote for Democratic candidates in the general election. Our treatment increased Republican primary turnout in this sample by 1.6 percentage points while reducing turnout in the Democratic Primary by .5 percentage points. Supplementing our experiment with surveys before and after the primary, we estimate that each vote cast by Democratic-leaning voters in the Republican primary had between a 78% and 95% probability of supporting the relatively moderate Republican primary candidate. We argue that voters are capable of sophisticated, risk-mitigating behavior in primaries.

Ahead of the 2024 election, Democratic and Republican voters expressed not merely discontent at the policy priorities of the opposing party's candidate for president, but alarm at the risks they posed to democracy. But this alarm about politicians' commitment (or lack thereof) to democratic norms did not transform the behavior of most voters. A growing body of research shows that warnings of democratic threat are not especially persuasive (Walk et al., 2024), that observed antidemocratic behavior causes only moderate electoral sanctioning (Graham & Svobik, 2020), and that citizens are often able to convince themselves of the acceptability of antidemocratic behaviors (Krishnarajan, 2023). More generally, Americans across party lines are ever more willing to tolerate misbehavior from co-partisan candidates. In an era of few "floating voters" (Smidt, 2017), it is improbable

that partisans of one party will sanction their own candidate's misbehaviors.

But in states with open or semi-open primaries, the 2024 presidential primaries offered a rare opportunity for voters to affect outcomes without defecting from their partisan commitments in general elections. Specifically, self-identified Democrats in open primary states could "crossover vote" by turning out to the Republican primary and voting against Donald Trump, a candidate whom many feared was "radical...seeking revenge" with ambitions to be a "fascist" or "king."¹ Drawing on the typology proposed by Cho and Kang (2015), we refer to this behavior as "hedging," or turning out in the opposing party's primary to support the out-party candidate you genuinely prefer, thereby lessening the cost of the opposing party winning the general election. This is in contrast to "raiding," or turning out in the opposing party's primary to support an out-party candidate you do not prefer, in hopes that nominating them will give your party a better chance of winning the general election. By crossover voting and hedging, Democratic voters could reduce the

Verification Materials: The materials required to verify the computational reproducibility of the results, procedures, and analyses in this article are available on the *American Journal of Political Science* Dataverse within the Harvard Dataverse Network, at: <https://doi.org/10.7910/DVN/NCONLM>

The Cornell Center for Social Sciences verified that the data and replication code submitted to the AJPS Dataverse replicates the numerical results reported in the main text of this article.

¹ These quotes are from open-ended responses to surveys conducted by our partner organization of self-identified Democrats in open primary states.

cost of a Republican candidate winning the general. Republicans have similar opportunities to vote against extremist Democrats, though not during the mostly uncontested 2024 Democratic presidential primaries.

However, this “crossover voting” is costly and requires surmounting multiple obstacles. Informational barriers are high. Would-be crossover voters must understand eligibility rules and follow multiple primaries in parallel. Psychological barriers are also considerable. Voters who vote against the opposing party’s extreme candidates might believe they are harming their own party’s general election chances (Cho & Kang, 2015), in line with evidence that extremist or antidemocratic candidates perform worse in general elections (Bartels & Carnes, 2023; Graham & Svobik, 2020). Would-be crossover voters must also reconcile with social norms that may discourage an unusual turnout decision (Gerber, Green, & Larimer, 2008; Thornburg, 2023).

We report the results of a first-of-its-kind field experiment conducted by a political action committee (PAC) that sought to increase the rate of crossover voting during the “semi-open” 2024 New Hampshire Republican presidential primary in order to increase the vote share of Nikki Haley. Specifically, we explore the effectiveness of efforts to encourage crossover voting in the Republican primary among individuals who were registered as undeclared² and modeled as likely to support Joe Biden in the general election. We find that encouragements to crossover vote increase turnout of Democratic-leaning voters to the Republican primary by 1.6 percentage points while reducing turnout to the Democratic primary by .5 percentage points. Supplementing these data with pre- and post-election surveys that ask Democratic-leaning voters’ preferences conditional on crossover voting, we predict that the vast majority of these crossover voters supported the more moderate candidate in the Republican primary. Exploring preregistered heterogeneous effects, we find that treatment is not differentially effective based on voters’ individual-level characteristics, but voters who live with registered partisans are somewhat less responsive to treatment than voters who do not. We also find, surprisingly, that encouragements to crossover vote can demobilize registered Republicans when they live with voters in our experimental universe.

As voters increasingly fear the opposing party winning office, we show that voters can use an unconventional avenue to lower the stakes of general elections. Near-uniformly, the voters we study prefer that their party face a moderate candidate in the general, even at the cost of their party’s chances

of winning. This is particularly impressive given three dynamics: Voters in other contexts fail to take steps to protect democracy even when they express concerns about it, partisan elites often encourage voters to support more extreme opposing party primary candidates to undermine the opposing party’s chances, and the 2024 election was a time of high partisan animosity.³ In all, we demonstrate that when electoral institutions allow for crossover voting, voters make a choice to (as they see it) protect democracy and hedge against adverse outcomes even at the expense of their party’s electoral prospects. These mechanisms offer a path forward for partisans across party lines to reduce the often-escalating stakes of general elections and improve the chances of moderate candidates.

A THEORY ON CROSSOVER VOTING

We view the choice to crossover vote as incorporating logic from both standard rational choice accounts of voting (Aldrich, 1993; Riker & Ordeshook, 1968) and strategic voting models in which voters trade-off instrumental and sincere motivations. Our experimental universe—which was eligible to vote in either party’s primary—faces two decisions related to crossover voting: first, whether to turn out to their own party’s primary,⁴ the opposing party’s primary, or abstain; and second, conditional on turning out to the opposing party’s primary, whether to hedge by voting for their most preferred opposing party candidate, or to raid by voting for the least electable opposing party candidate.

We begin with the second choice, vote choice in the opposing party’s primary, in part because this preference is an input into the calculus of turnout. This choice, for self-identified Democrats and supporters of Joe Biden, required trading off the benefits of a sincerely preferred candidate winning the Republican primary and reducing the risks of a Republican general election victory (hedging) against the benefits of an ideologically extreme candidate winning the Republican primary and increasing the Democratic nominee’s chances in the general election (raiding). Beyond these strategic concerns, voters must also weigh normative objections to raiding (Hussein, Lee, & Wheeler, 2025, shows voters disapprove of advertisements to “meddle” in the opposing party’s primary by promoting more extreme candidates). As an empirical matter, the pre-election surveys covered later demonstrate that would-be Democratic crossover voters overwhelmingly support Nikki Haley, the most popular moderate candidate, conditional on turnout to the Republican

² New Hampshire voters can register with a party or as undeclared. Undeclared voters who vote in a party primary are registered as a member of that party, but can fill out a card when leaving polling stations to maintain their undeclared registration.

³ <https://polarizationresearchlab.org/2024/10/30/report-path-to-2024-election-expectations/>

⁴ Because New Hampshire’s semi-open primaries allow only “undeclared” voters to choose which primary to vote in, we understand partisanship as both a voter’s self-identification and their support for a party in the general election.

primary, and disapprove of raiding. These results suggest that even if some Biden voters decided that the risks of a Trump presidency were worth decreasing Republicans' perceived general election odds, raiding-specific normative concerns overwhelm any strategic impetus to raid.

Conditional on this candidate choice, voters' decision to turn out to their own party's primary, the opposing party's primary, or abstain can be conceptualized as a modified version of the standard turnout model (Aldrich & Jenke, 2017; Gerber, Green, & Larimer, 2008) in which each primary produces unique costs and benefits. Differences in primaries' competitiveness, a voters' own pivotality, and the psychological non-material "duty" implications of turnout influence these costs and benefits.

But at least three barriers stand between partisans and crossover voting. First, voters face informational challenges to crossover voting. They must know they are eligible to crossover vote, and be familiar with the opposing party's primary candidates (a task harder than familiarity with their own party's primary, which Gerber et al., 2017, show voters struggle with). If opposing a front runner, crossover voters must also coordinate on an alternative. In the case of the New Hampshire presidential primary, this challenge may have been accentuated by the changing slate of candidates (Bartels, 1989), but mitigated by heavy media coverage and campaign spending.

Second, voters may have trouble weighing the relative pivotality of primaries. Past descriptive literature finds that voters account for competitiveness of elections when they are eligible to vote in multiple mutually exclusive elections (Dahl et al., 2023; Thornburg, 2023), while other experiments find null effects of emphasizing the closeness of an election on turnout (Enos & Fowler, 2014; Gerber et al., 2020).

Finally, voters "duty" payoffs can pull voters in either direction. Voters may derive particular duty benefits from voting in their own party's primary because it affirms their partisan identity, or costs if the opposing party's primary is the only competitive primary and a vote in their party's primary is perceived as wasted or ineffective. Voters could also derive benefits from voting in the opposing party's primary, because they feel their vote "counts," or is more effective at preventing extremism, or face costs if others disapprove of crossover voting or if a voter views any type of strategic voting as normatively problematic (Gerber et al., 2017, argues voters hold "exclusionary" views of who should vote in primaries). This normative consideration is shaped by whether a voter hedges or raids as discussed above.

The intervention studied in this paper aimed to change several inputs into this calculus of voting and lower the barriers discussed above. First, by informing undeclared voters that they were eligible to vote in the

Republican primary, our partner organization sought to lower informational barriers to crossover voting. Second, emphasizing that the Democratic primary is not competitive and that the Republican primary candidates differ on many policies emphasize both the higher pivotality and stakes of the Republican primary. Third, portions of treatments that named Nikki Haley help would-be crossover voters coordinate on an alternative to Donald Trump. Finally, using a social pressure message specific to the Republican primary orients the duty motivations of voters toward crossing over, and specifically toward hedging. Our partner organization capitalized on a strong legacy of crossover voting in New Hampshire, which may have further blunted normative social objections to crossover voting (McCarthy & Tukey, 1978).⁵

ETHICAL CONSIDERATIONS

Through the course of working on this project, periodic media coverage and academics raised ethical concerns that crossover voting, encouraging crossover voting, and studying this encouragement constitute underhanded electioneering. We believe that is not the case for a number of reasons.

First, political elites (PACs and party campaign committees) are engaged in efforts to both encourage crossover voting and further the chances of particular opposing party candidates. Some, like the effort studied here, encourage crossover voting to further moderate and perceived pro-democratic candidates. Others constitute partisan efforts to "raid" opposing party's primaries by supporting extremists who are expected to perform worse in the general election. In a non-exhaustive audit of these efforts in the 2022 elections, we found 17 distinct efforts that spent over \$52 million. Political science is interested in the activity of political donors and PACs, and the results of their spending on candidate positioning, election outcomes, and representation (Kalla & Broockman, 2015; Masket & Shor, 2015; Skocpol & Hertel-Fernandez, 2016; Schatzinger & Martin, 2020). Here, we examine increasingly frequent PAC efforts aimed at the opposing party's primary to obtain a fuller picture of PAC activity today.

Second, increasingly popular institutional reforms of primaries (e.g., semi- and fully open primaries) not only legalize crossover voting, but rely on crossover voting to realize their proposed benefits. Those who advocate for open primaries argue that open primaries will yield more moderate candidates precisely because they allow a wider segment of voters to participate in a primary (Boatright, 2024). But these benefits

⁵ Reflecting this legacy, 41% of our control group appear to crossover vote even without experimental encouragements.

depend on a form of hedging: voters who do not identify with a party supporting a moderate candidate in that party's primary. Though scholarship has struggled to find strong effects of open primary reforms, this might be because there is a low level of crossover voting in the absence of encouragements like the one we assess here. Studying crossover voting is essential to understanding the effects of primary election institutional reform.

Third, as the number of truly competitive general elections decreases, there is a large segment of effectively disenfranchised American voters: voters who identify with the minority party in their district or state. For example, a Republican voter in a solidly Democratic state or congressional district where the Democratic nominee is all but assured victory in the general election does not vote in the one truly competitive election: the Democratic primary. When open primary institutions allow for crossover voting, these voters have the opportunity to cast a ballot in the election that decides their officials. As scholarship has found elected officials especially responsive to primary voters, crossover voting allows these voters a better chance at representative and responsive officials (Anderson, Butler, & Harbridge-Yong, 2020).

Finally, the choice a crossover voter makes between "raiding" and "hedging" is comparable to other types of strategic voting. Downs (1957) notes that, in two-party systems, voters who support third-party candidates who voters know have little chance of winning may engage in two types of voting (49): supporting the most ideologically proximate of the major candidates, or casting a sincere vote for the nonviable candidate in hope of affecting the major party's incentives moving forward. Political science has explored questions about strategic voting because the incidence of strategic voting affects our understanding of electoral institutions, election outcomes, and voter capacity (Abramson et al., 2010; Blais & Nadeau, 1996; Duverger, 1954; Duch & Palmer, 2002; Eggers & Vivyan, 2020; Lijphart, 1994). Understanding vote choice among crossover voters will likewise inform our beliefs on these topics.

We took further steps to ensure our research was conducted within ethical bounds. We studied an intervention that was already in progress, worked with our partner organization to randomize only which voters were targeted (rather than picking who was eligible to be targeted), and examined the effects of an intervention designed to increase hedging in favor of a moderate candidate.

HYPOTHESES

Our theoretical expectations lead us to the following preregistered hypotheses regarding the effectiveness

of a modified "get out the vote" (GOTV) intervention to encourage crossover turnout.

First, we expect our partner organization's treatment to be effective in increasing crossover voting. GOTV interventions are generally effective across many election types (Green & Gerber, 2019), and this intervention was designed to mitigate the specific challenges to crossover voting. What's more, the electoral environment of the New Hampshire Republican presidential primary (an uncompetitive Democratic primary and an opposing party primary candidate perceived as very threatening) contains strong motivations for Democratic-leaning voters to cross over.

Hypothesis 1. Encouragement to crossover vote will increase turnout in the Republican primary.

The peculiar mechanics of crossover voting also lead us to expect that these treatments will yield a reduction in turnout in the Democratic primary because our treatment emphasizes the futility of participation in that election. Few papers have studied contexts where voters choose not only whether to turnout but also in which of several mutually exclusive elections to vote. But Dahl et al. (2023) and Thornburg (2023) suggest that voters choosing between multiple mutually exclusive elections consider relative pivotality and shift votes away from the least competitive context. This logic suggests that, in our experimental context, voters who had previously intended to vote in the Democratic primary would shift their turnout decision to the Republican primary.

Hypothesis 2. Encouragement to crossover vote will decrease turnout in the Democratic primary.

Past literature suggests that education may help voters mitigate the informational and strategic reasoning challenges to crossover voting (Duch & Palmer, 2002; Eggers & Vivyan, 2020; Miranda & Meloni, 2016). As a result, we expect educated voters to be more responsive to encouragements to crossover vote.

Hypothesis 3. The conditional average treatment effect (CATE) of encouragement to crossover vote will be larger for voters with higher modeled education scores.

Similarly, we expect voters who are more engaged with politics to be differentially responsive to treatment. They could be more responsive, as the informational barriers to crossing over are lower because they are more familiar with electoral rules and the opposing party's primary candidates, or less responsive, if their past turnout indicates a stronger loyalty to the Democratic party or they have stronger normative opinions about who should vote in primaries.

Hypothesis 4. The CATE of encouragement to crossover vote will be different for voters who have turned out to vote in recent Democratic primaries.

We also hypothesize how treatment will effect voters outside our experimental universe but who live with voters who are part of our sample. Specifically, partisan registrants who are only eligible to vote in their own party's primary but who are exposed to spillover treatment by their housemates may encounter messaging about how much more important the Republican primary is than the Democratic primary. These voters will similarly weigh the pivotality of their primary in their turnout calculus to turn out or abstain. As a result, we predict that the informational components of treatment will change the turnout behavior of partisan housemates.

Hypothesis 5. Partisan registered voters indirectly treated through within-household spillovers will, if Republican, vote more in the Republican primary and, if Democratic, vote less in the Democratic primary.

Finally, we have two hypotheses we did not pre-register. First, we predict that voters with different modeled partisanship scores will react differently to treatment, because they will have different psychological “duty” payoffs to crossover voting or voting in their own party's primary. We have no prior about whether those with a higher modeled likelihood of supporting Democrats in general elections will be more or less affected by treatment. More “Democratic” voters may be more responsive to treatment because they feel more threatened by perceived threats to democracy and extremism and so are more motivated to oppose Trump. But more “Democratic” voters may also be less responsive to treatment because they gain stronger affective benefits from participating in their own party and so are less willing to forgo their vote in the Democratic primary.

Hypothesis 6. The CATE of encouragement to crossover vote will be different for voters who have a higher modeled likelihood of supporting Democrats.

Second, we predict that voters in our experimental universe who live with partisan housemates will respond to our treatment differently. These voters will also have “duty” payoffs to crossover voting, as living with a registered partisan will lead to different norms around the acceptability of crossover voting. For voters living with registered Democrats, their partisan housemate may encourage their crossover voting to prevent extremism. But for voters living with registered Republicans, their partisan housemate may discourage their crossover voting because they view crossing over as meddling in their own primary. As such, we

predict voters living with Democratic housemates will be more responsive to treatment while voters living with Republican housemates will be less responsive to treatment.

Hypothesis 7. The CATE of encouragement to crossover vote will be larger for voters living with a registered Democrat, while the CATE will be smaller for voters living with a registered Republican.

METHOD

To investigate whether primary voters can be induced to strategically cross over, we carried out a preregistered randomized controlled trial in January of 2024 in the New Hampshire Republican presidential primary. The treatment was designed and administered by our partner organization, a PAC that works to promote the nomination of moderate primary candidates. In addition, we examine two sources of public opinion data: first, three survey experiments carried out by our partner organization in the weeks before the field experiment—these experiments helped the PAC select a treatment for the field experiment, and, for our purposes, clarified the vote choice and motivation of voters who choose to cross over; second, a follow-up survey conducted by our partner organization after the election among those in our field experimental sample who did crossover vote—this survey helps us to confirm that our interpretation of administrative data accurately described crossover voting.

Pre-election surveys

We first investigated the mechanics of crossover voting in the presidential primary by reviewing the results of three survey experiments (total $N = 4,381$) conducted on Lucid between December 26, 2023, and January 3, 2024, by a private polling firm hired by our partner organization. These experiments were designed to message test approaches to encourage crossover voting in the New Hampshire primary and used quota sampling to roughly match the demographics of New Hampshire undeclared voters on education and race from a pool of self-identified Democrats or independents who resided in states with open primaries (a population facing similar primary crossover voting trade-offs). For each experiment, respondents were asked pretreatment demographic and attitudinal questions, randomly assigned to see one of several treatments or a placebo using simple random assignment, and then asked outcome questions: Which primary they would turnout to, Republican primary vote choice, and approval of hedging behavior. Because many of the treatment arms had only

minor variations in language (described fully in the Online Appendix, p. 18), we summarize as follows: Experiment 1 independently randomized several features of each message, including naming Nikki Haley, describing her polling gains against Donald Trump, and telling respondents not to waste their vote in the Democratic primary. Experiment 2 tested messages about Trump trying to restrict undeclared voters' ability to vote in the primary against messages about Haley's differences from Trump on important issues, including commitment to democracy. Experiment 3 tested an affective message priming dislike of Trump against a condition which made the case that Trump threatens democracy.

Because we did not conduct these experiments, the results were not preregistered so we restrict our main analyses to the outcomes of interest: turnout intention, candidate choice in the Republican primary, and approval of hedging behavior writ large. For simplicity, we pool the treatments in each experiment and compare them to the control.⁶ Finally, we keep our covariates in line with those preregistered for the field experiment (all of which are available on the voter file): gender, race, education, partisanship, and prior turnout.

Field experiment

New Hampshire has semi-open primary elections, such that unaffiliated voters (or voters who are registered to vote but not registered for any political party) can choose whether to vote in the Democratic or Republican primary. Our experimental universe is the 83,902 undeclared New Hampshire voters who have available cell phone numbers, have voted in a Democratic, but not a Republican, presidential primary since 2008 (inclusively), have a partisanship score⁷ of above 40,⁸ and are in a household with fewer than five undeclared voters.⁹ Consistent with the state's broader demographics, the sample was 97% White and 56% female.¹⁰ As designed, modeled partisanship leaned heavily toward Democratic voters, with a mean "VCI" of 85 and a median of 90. Modeled education is more evenly distributed with a median of 43 and a mean of 44.¹¹ Both distributions are shown in Figure 1. Our

main outcome of interest is turnout to the Republican primary, which we measure using the voter file.¹²

Treatment was cluster-randomized at the household level. The 83,902 voters in our sample lived in 72,191 households, 75% of which were assigned treated and 25% assigned to the control. The treatment itself consisted of seven text messages,¹³ seven social media impressions served to all individuals within a treated household on Meta-owned social media platforms, and three mailers encouraging voters in the treated households to crossover vote.¹⁴

While all voters in a treated household received texts and social media ads, only one undeclared voter in each household had mailers directly addressed to them. As such, a first round of assignment placed households in treatment and control, and a second round of assignment randomly selected one voter from the voters in each treated household to receive mailers. This process meant that individuals in households with only one voter in our experimental universe were eligible for treatment via texts, social media ads and mailers ("Mail + Digital treatment"), or control. In contrast, individuals in treated households with multiple voters in our universe could be assigned to receive the same "Mail + Digital" treatment, with mailers addressed to them, or treatment with only texts and social media ads ("Digital Only" treatment) if they were not personally assigned to receive mailers. The probability of assignment to each condition for voters in a household with h_n experimental subjects is $\frac{3}{4} * \frac{1}{h_n}$ for "Mail + Digital" treatment, $\frac{3}{4} * \frac{h_n-1}{h_n}$ for treatment via only social media and texts, and $\frac{1}{4}$ for pure control. The results of these equations and the probability of assignment to each condition for voters in each household size are shown in Table 1. In total, 54,141 households with 62,932 individuals were treated with texts and social media ads, and 54,141 of those same individuals (1 per treated households) also received mailers directly. In all 18,050 households with 20,970 individuals were placed in control and were not contacted. Table 1 illustrates the eligibility and distribution of treatment.

In addition, we block randomized on whether the household was also being contacted by another organization, which encouraged Republican primary turnout in general terms.¹⁵ We did not block on cluster size (i.e., how many voters in our experimental

⁶ Treatment effects for each treatment condition are shown in the Online Appendix, p. 21.

⁷ Referred to as Vote Choice Index or "VCI": a Catalist constructed measure of a voter's likelihood of supporting the Democratic candidate in the general election.

⁸ A voter is $\geq 40\%$ likely to vote for a Democrat in the next general election.

⁹ We also remove individuals that are targeted by another contacting effort by our partner organization, but leave those undeclared voters not personally targeted but living in households targeted by the other effort.

¹⁰ Full balances tests are included in the Online Appendix, p. 8.

¹¹ This variable can be interpreted as an individual's likelihood of having completed a 4-year college degree.

¹² The pre-analysis plan for this experiment is available [here](#).

¹³ Our partner organization estimated a low successful delivery rate, which caused them to expand the intensity of text treatments from a planned 4 to 7.

¹⁴ Online Appendix, p. 5, shows all treatment material.

¹⁵ During the course of the experiment, we became aware that an unrelated organization expended significant resources encouraging crossover voting across the state. We are not aware of their specific targeting criteria but our sample is sufficiently large that if any large subset of our universe was exposed to treatment, exposure should be evenly balanced between treatment and control groups.

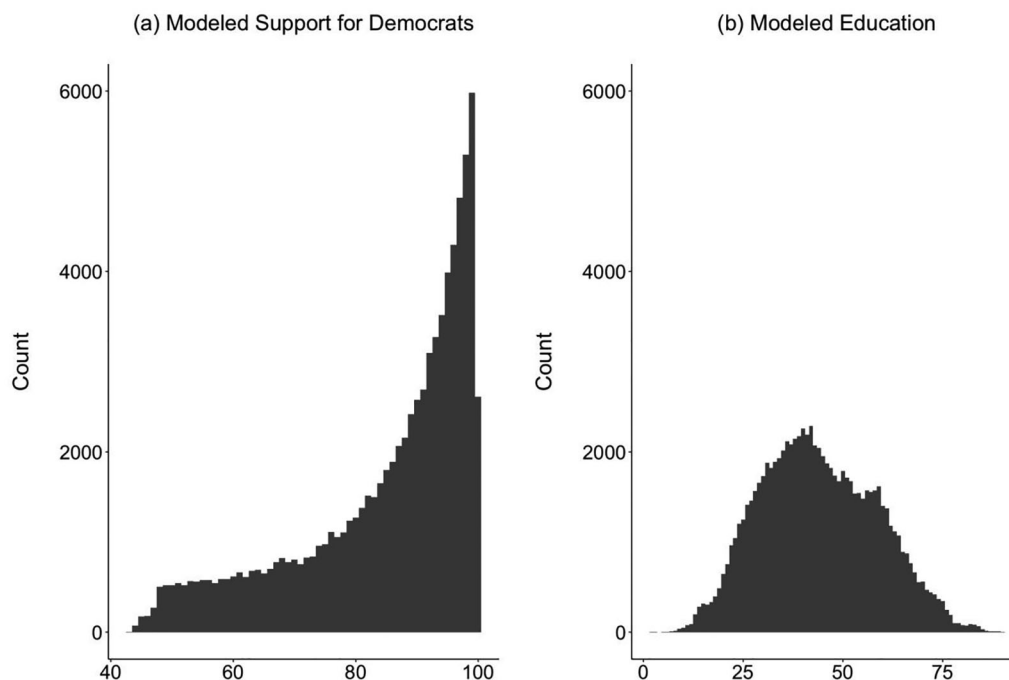


FIGURE 1 Distribution of modeled support for Democrats and education. *Notes:* This figure shows a histogram of modeled support for Democrats in subplot (a), and of modeled education in subplot (b), in our sample.

TABLE 1 Treatment assignment probability for a voter in differently sized households.

Household size	Mail + digital (mailer, social media ads, texts)	Digital only (social media ads, texts, spillover to mailers)	Control
One voter	.750	.000	.250
Two voters	.375	.375	.250
Three voters	.250	.500	.250
Four voters	.188	.562	.250

universe a household had). Though random assignment of clusters of unequal size creates the potential for small-sample bias (Gerber & Green, 2012; Middleton & Aronow, 2015), our study has a sufficient number of clusters such that this bias approaches zero.¹⁶

As discussed in the theory section, this experiment, as the first of its kind, aimed to encourage crossover voting through as many avenues as possible. We view the bundled treatment as encouraging crossover voting through (1) informing voters of their eligibility to cross over, reducing informational barriers; (2) shifting beliefs about competitiveness of the primaries to encourage voters to weight the relative pivotality of the two primaries; (3) priming primary-specific social duty concerns to increase the acceptability of crossing over; (4) emphasizing the unacceptability of

Trump to Democratic voters through both ideological and affective mechanisms, thereby emphasizing the policy benefits to crossing over; and (4) naming Nikki Haley as the ‘correct’ out-party candidate to support, mitigating informational barriers to learning about the opposing party’s primary candidates and reducing the difficulty of coordination among voters seeking a moderate candidate.

While mailers were successfully delivered to all treated individuals, texts and social media ads did not reach every voter assigned to treatment. Specifically, 82% ($\frac{51,373}{62,983}$) of individuals assigned to receive texts had valid phone numbers and were sent texts. An unknown number of these voters did not receive the first four out of seven texts, because some spam filters blocked pictures with embedded hyperlinks. As such, we do not know the overall rate of compliance for text encouragements. For social media ads, only 30% ($\frac{19,052}{62,983}$) of individuals assigned to treatment

¹⁶ Estimators sub-setting to each cluster size are unbiased in finite samples and our results are robust to an analysis of only one-voter households in the Online Appendix, p. 14.

were successfully matched to accounts on Facebook or Instagram and received ads. For this reason, we estimate only the intent-to-treat (ITT) effect of treatment on overall turnout, turnout to the Democratic primary, and, our main outcome of interest, turnout to the Republican primary.¹⁷

We use two approaches to estimate the ITT. First, we assess the effect of treatment at the household-cluster level, with all units having the same $\frac{3}{4}$ odds of assignment to this treatment. Second, we explore the effects of individual-level assignment for different treatments: receiving encouragements via text, social media ads, and mail (“Mail + Digital” treatment), and receiving encouragements only via text and social media ads (“Digital Only” treatment). ITT estimates for the effect of “Digital Only” treatment also include any indirect effect of their housemates receiving mail encouragements, because those assigned to Digital Only treatment always had a housemate assigned to Mail + Digital treatment. Because individuals living in one-voter households had zero probability of assignment to Digital Only treatment, we separately estimate ITTs for individual-level treatment assignments for voters in one-voter households and multi-voter households.¹⁸ For all estimates, we cluster standard errors at the household level. Consistent with our pre-analysis plan, all models control for LASSO-selected covariates¹⁹ drawn from the voter file.²⁰

In multi-voter households, individuals had varying odds of receiving the Mail + Digital, and Digital Only treatment depending on the size of their household. For models estimating the ITT of each level of treatment in multi-voter households, we account for these varying odds with inverse probability weighting (IPW) (Gerber & Green, 2012) based on the odds of assignment to each condition, shown in Table 1.

We explore heterogeneous effects by four variables: first, past turnout to the 2020 Democratic presidential primary, as recorded on the voter file; second and third, voters’ modeled level of education and support for Democrats in general elections, and fourth, by whether a voter lives with a registered

partisan voter by matching voters based on mailing address.²¹

Finally, a note on spillovers and treatment intensity. As noted above, our analyses of individual-level treatment serve to bundle direct (Digital Only or Mail + Digital) and indirect (the treatment status of other voters in the household) treatment effects. In contrast, we are unconcerned with between-household spillovers given the geographically diffuse nature of the sample and prior work showing negligible spillovers between households even in denser settings (Sinclair, McConnell, & Green, 2012). However, we use the term “spillover” to refer to individuals who were not part of the experimental universe and thus could not be directly treated but did live in households with experimental subjects (similarly to how Nickerson, 2008, uses the term). Specifically, we preregistered estimating the effect of living with treated voters on registered Republicans, registered Democrats, and undeclared voters who were eligible to vote in either primary but were excluded from our universe based on some other criteria. We cluster standard errors for spillover analyses at the household level.

Post-election survey

To confirm our interpretation of the administrative turnout data, our partner organization conducted a post-election survey that verified the primary vote choice of 370 experimental participants who crossed over. The survey was administered over the phone to the cellphone numbers of all experimental subjects in treatment or control who turned out to vote in the Republican primary, a total of 35,265 voters. Each voter was called once. Respondents were further matched to the voter file by first name, birth year, and gender. With a total of 370 responses, the survey had just over a 1% response rate, in line with the, on average, 2% response rate to phone polls making repeated calls that offer no compensation for responses.²² Though the survey was not designed to be perfectly representative of our experimental universe, respondents were distributed roughly evenly across treatment and control, and spanned the full range of modeled partisanship in the sample. Respondents were also broadly similar to the full set of voters who turned out in the Republican primary along modeled partisanship and education (though the sample was slightly less Democratic than modeled in their 2024 general election vote choice). We note that this survey provides insight into crossover behavior but does not isolate the vote

¹⁷ This focus on the ITT likely attenuates our estimates. Since we cannot estimate compliance for the full treatment, we are unable to estimate a complier-average-causal effect (CACE), though we briefly discuss how cell phone match probabilities help us to understand compliance in the Online Appendix, p. 12.

¹⁸ This is a deviation from our pre-analysis plan because the IPW analysis outlined in the PAP would not correctly adjust for differing probabilities of individual-level assignment if we analyzed all households together. Because voters in one-voter households had no chance of assignment to the “Digital Only” condition, any comparison between these subjects and subjects in multi-voter households is non-experimental.

¹⁹ We used LASSO to select among some covariates included in the voter file, while excluding minor covariates due to missingness.

²⁰ Known targeting by similar interventions, household size, gender, race, modeled Democratic vote likelihood (VCI), modeled education, state house and senate district, and prior turnout in the 2020 general election and 2020 Democratic primary.

²¹ Heterogeneous effects analysis of the first two variables (prior Democratic primary turnout and education) were preregistered, while heterogeneous effects by VCI and partisan housemates were not.

²² <https://www.nytimes.com/article/times-siena-poll-methodology.html#link-1aa612f8>

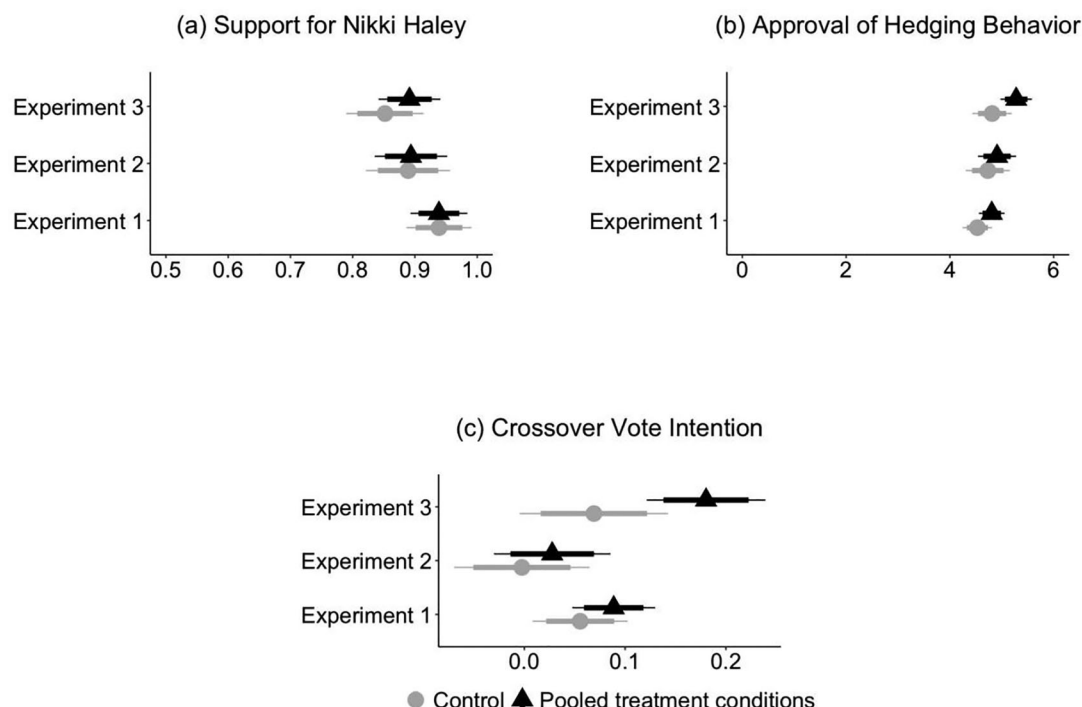


FIGURE 2 Treatment effects on (a) support for Haley, (b) approval of hedging behaviors, and (c) crossover vote intention. *Notes:* This figure shows that survey treatments increase support for the moderate candidate, approval of hedging, and intention to crossover vote.

choice of treatment-responsive voters (treated voters who crossed over but would have abstained or voted in the Democratic primary had they not been treated).

RESULTS

Pre-election surveys

We begin with the results of the survey experiments conducted by our partner organization before the election to inform the design of the field experiment. These experiments allowed our partner organization to select the most effective encouragement (namely, those that primed affective dislike for Trump and used social pressure) for the succeeding field experiment, and demonstrated voters' propensity to hedge (and distaste for raiding) in their support for the moderate Republican primary candidate. For each of three experiments, we pool all treatments and show results on three outcomes of interest: support for the candidate portrayed by treatments as a pro-democracy moderate (Figure 2a), support for hedging behavior (Figure 2b), and intention to crossover vote (Figure 2c).²³

Figure 2a shows that, conditional on voting in the Republican primary, Democratic-leaning voters in open primary states almost unanimously support

Nikki Haley over Donald Trump. Treatments that our partner organization designed to emphasize Trump's antidemocratic behavior had little effect on support, because more than 90% of respondents in control already supported Haley across three samples. These results illustrate that would-be crossover voters exhibit widespread agreement on their preferred candidate.

Analyzing the open-ended responses that ask respondents who would vote against Trump in the primary and support Biden in the general to explain their primary vote choice suggests that preventing extremism and threats to democracy are key motivations.²⁴ In all 7.4% of respondents mention words around "danger" and "violence," and 5.5% mention concerns for democracy, like "autocrat" and "dictatorship"; 5.4% are concerned with ideology, using words like "moderate" and "extremist." Experimental subjects who vote against Trump do so because of their ideological reasons and concerns about democracy.

Figure 2b shows mean approval of hedging behavior, by asking respondents placed in treatment and control for approval (on a 7-point scale) of "Democrats and Independents voting in the Republican presidential primary against extremist candidates." Higher values indicate stronger approval. In both treatment and control, voters approve of hedging behavior. Figure 2c shows that treatments encouraging crossover voting

²³ Online Appendix, p. 20, shows treatment-specific results.

²⁴ Exact terms categorized as relevant to democracy, threat, and extremism can be found in the Online Appendix, p. 17.

TABLE 2 Treatment effects for household-level assignment.

	Republican primary turnout	Democratic primary turnout	Overall turnout
Treated	.016** (.004)	-.005 [†] (.003)	.011** (.004)
Observations	83,740	83,740	83,740
R ²	.071	.027	.096

Notes: Models include demographic covariates and cluster standard errors by household.

[†] $p < .10$, * $p < .05$, ** $p < .01$.

do increase voters' willingness to cross over. As the survey treatments iterate, this effect becomes statistically significant.

Beyond informing our partner organization's treatment for the succeeding field experiment, we derive three stylized facts from this survey data. First, vote choice in a two-way primary ballot suggests our sample generates votes for Nikki Haley at a high rate, even in the control group. Second, our sample is more comfortable with hedging and associated behaviors than raiding, even in the absence of specific encouragements. Pooling across the Biden supporters in the control groups ($N = 674$), respondents supported hedging at an average of 4.7 on a 7-point scale, while they supported raiding at an average of 3.2 on the same scale. Third, a variety of treatments seem effective at increasing willingness to crossover vote, especially the treatments in Experiment 3.

Field experimental results

In the New Hampshire presidential primary, turnout among our control group was 53.0%, including 40.8% turnout in the Republican primary and 11.1% turnout in the Democratic primary. We believe high baseline turnout to the Republican primary results both from broader campaign efforts to encourage crossover voting and heavy media coverage in New Hampshire that emphasized the possibility of crossover voting.

In Table 2, we report the average treatment effect of receiving any encouragement to crossover vote compared to receiving no such encouragement. In other words, we compare treated to untreated households. As discussed, these results pool over different levels of treatment.

We find that pooled treatment increases turnout in the Republican primary by a statistically significant 1.6 percentage points, and decreases turnout in the Democratic primary by a marginally significant .5 percentage points. It also significantly increases overall primary turnout by 1.1 percentage points.²⁵

These results are consistent with our preregistered Hypotheses 1 and 2.

Next, we assess the *levels* of treatment in Tables 3 and 4. As discussed earlier, we estimate the effects of individual-level treatment separately for households with one voter in our experimental universe and those with more than one voter.

We find that our overall treatment effect is driven by those who received Mail + Digital treatment. Table 3 shows that in one-voter households, Mail + Digital treatment caused a significant 1.7 percentage point increase in turnout to the Republican primary, similar to the increase found when pooling across levels of treatment. Mail + Digital treatment also significantly decreased turnout to the Democratic primary by .5 percentage points and increased overall turnout by 1.2 percentage points, identical to the pooled analysis estimates. Table 4 shows that in multi-voter households, the estimated effects of Mail + Digital treatment are nearly identical (increasing Republican primary turnout by 1.7 percentage points and decreasing Democratic primary turnout by .8 percentage points), though the estimates are much less precise because most of our sample lives in households with only one voter in our experimental universe.

However, in multi-voter households, digital only treatment does not significantly change turnout. Though the direction of the effects is consistent with our priors and the effects of those treated with mail, social media, and texts, the effects of treatment through only texts and social media are smaller and fall short of statistical significance. This gap may be accentuated because texts and social media were subject to noncompliance, while mailers were not. The insignificant effects of digital only treatment also suggest that the spillover effect of mailers sent to a housemate is small for voters placed in digital only treatment.

Because treatment not only increased Republican primary turnout but also decreased Democratic primary turnout, the Republican primary votes this experiment generated did not come solely from non-voters. Specifically, about $\frac{5}{8}$ of the Republican primary votes we generated appear to have come from if-untreated abstainers while $\frac{3}{8}$ come from if-untreated

²⁵ These results are robust to other model specifications in the Online Appendix, p. 2.

TABLE 3 Treatment effects of individual assignment, one-voter households.

	Republican primary turnout	Democratic primary turnout	Overall turnout
Mail + digital treatment	.017** (.004)	-.005 [†] (.003)	.012** (.004)
Observations	61,234	61,234	61,234
R ²	.071	.031	.095

Notes: Models include demographic covariates and cluster standard errors by household.

[†] $p < .10$, * $p < .05$, ** $p < .01$.

TABLE 4 Treatment effects of individual assignment, 2+ voter households.

	Republican primary turnout	Democratic primary turnout	Overall turnout
Mail + digital treatment	.017 [†] (.010)	-.008 (.006)	.009 (.009)
Digital only treatment	.010 (.010)	.000 (.006)	.010 (.009)
Observations	22,506	22,506	22,506
R ²	.074	.029	.093

Notes: Models include demographic covariates and cluster standard errors by household.

[†] $p < .10$, * $p < .05$, ** $p < .01$.

Democratic primary voters. Given the relatively low turnout in the Democratic primary in our control group, we estimate that treatment decreased Democratic turnout by a noticeably fraction of the total (nearly $\frac{1}{20}$ of the 11% of our control group who voted in the Democratic primary).

Heterogeneous effects of voter characteristics

Next, we explore treatment heterogeneity by preregistered variables (modeled education and prior primary turnout) and by modeled partisanship.

Table 5 shows the estimated ITT of receiving any household treatment on voters at different likelihoods of holding a college degree. The education variable is standardized such that a unit increase corresponds to a 10% increase in likelihood of holding a college degree. As a covariate, education correlates negatively with crossover voting, turning out to the Democratic primary, and turnout in general within the control group. Though contrary to the usual positive association between education and political engagement, educated voters in our sample may be more responsive to concerns about pivotality (as Eggers & Vivyan, 2020, suggest), and thus demobilized by the uncompetitiveness of the Democratic primary. Our preregistered expectation, though, was that educated voters would be more responsive to treatment because of

this greater appetite for strategic trade-offs and lower informational barriers. However, we do not find this to be the case; our treatment is no more effective among voters with higher education scores. However, more granular methods of estimating these heterogeneous effects suggest slight but insignificant increases in the treatment effect over large changes in likelihood of holding a college degree.²⁶

Second, we explore heterogeneity by partisanship. Table 6 shows the estimated ITT of receiving any treatment at the household level on all outcome variables for voters at different levels of “VCI,” or likelihood to support the Democratic candidate in a general election. As with education, VCI is standardized such that a one unit increase of the VCI variable corresponds to a 10% increased likelihood of supporting the Democrat. Across all models, we find that VCI operates as expected; a higher VCI score (or a higher likelihood of supporting a Democrat in the general election) is positively correlated to turning out in the Democratic primary but negatively correlated to turning out in the Republican primary. At baseline, voter behavior validates that VCI is a measure of partisanship; more Democratic voters vote in the Democratic, not Republican, primary.

We expected to see treatment efficacy vary by level of partisanship, as many of the payoffs and costs of

²⁶ Preregistered Generalized Random Forest analyses are displayed in the Online Appendix, p. 10, along with binned models.

TABLE 5 Heterogeneous effects by education score, household-level assignment.

	Republican primary turnout	Democratic primary turnout	Overall turnout
Treated	.016** (.004)	-.005 [†] (.003)	.016** (.004)
Education score	-.013** (.003)	-.011** (.002)	-.013** (.003)
Treated × Educ score	.004 (.003)	.002 (.002)	.004 (.003)
Observations	83,740	83,740	83,740
R ²	.071	.027	.071

Notes: Models include demographic covariates and cluster standard errors by household.

[†] $p < .10$, * $p < .05$, ** $p < .01$.

TABLE 6 Heterogeneous effects by Vote Choice Index (VCI), household-level assignment.

	Republican primary turnout	Democratic primary turnout	Overall turnout
Treated	.016** (.004)	-.005 [†] (.003)	.011** (.004)
Vote Choice Index	-.040** (.002)	.012** (.002)	-.028** (.002)
Treated × VCI	.001 (.003)	.001 (.002)	.002 (.003)
Observations	83,740	83,740	83,740
R ²	.071	.027	.096

Notes: Models include demographic covariates and cluster standard errors by household.

[†] $p < .10$, * $p < .05$, ** $p < .01$.

crossover voting tie into a voter's loyalty to the party. Treatment may be less effective among more Democratic voters because they derive a stronger affective benefit to voting in their party's primary or because they find voting for any Republican candidate distasteful. Or treatment may be more effective among more Democratic voters because they are more motivated to prevent Trump's nomination.

Instead, when interacted with household-level treatment in Table 6, we find no significant heterogeneity of treatment effect by voters' partisanship. This null result is robust to more sophisticated estimation strategies.²⁷ We note, however, that our power to detect heterogeneity along this dimension is limited given the by-design tight distribution of VCI in our sample.²⁸

Third, we explore how treatment effects differ between voters who voted in the last Democratic presidential primary and those who did not as another test of how partisanship modifies treatment efficacy. While

this analysis was preregistered, we had competing expectations of the direction of effect.

Again, and contrary to our expectations, we find no heterogeneous effect of treatment by voters' prior Democratic primary engagement. Table 7 shows small and insignificant differences in treatment effects on voters who did and did not vote in the 2020 Democratic primary. This finding could be the result of averaging out a positive and negative heterogeneous effect among engaged past Democratic voters who are motivated to stop Trump, and engaged past Democratic voters who will not vote for any Republican. Still, voters with a recent record of engagement in Democratic primaries do not appear to react differently to treatment.

Heterogeneous effects by voters' social context

We now examine how the social context of voters, as distinct from their demographic characteristics, shapes their openness to crossover voting. While not

²⁷ General Random Forest, LOESS, and binned analyses can be found in the Online Appendix, p. 10.

²⁸ We discuss this further in the Online Appendix, p. 23.

TABLE 7 Heterogenous effects by 2020 Democratic primary turnout, household-level assignment.

	Republican primary turnout	Democratic primary turnout	Overall turnout
Treated	.014* (.007)	-.006 [†] (.003)	.008 (.007)
Prior Dem vote	.158** (.007)	.071** (.004)	.228** (.007)
Treated × Prior Dem vote	.003 (.008)	.002 (.005)	.005 (.008)
Observations	83,740	83,740	83,740
R ²	.071	.027	.096

Notes: Models include demographic covariates and cluster standard errors by household.

[†] $p < .10$, * $p < .05$, ** $p < .01$.

TABLE 8 Heterogeneous effects by registration status of housemates.

	Republican primary turnout	Democratic primary turnout	Overall turnout
Treated	.020** (.005)	.020** (.005)	.017** (.004)
Republican in house	-.017 (.012)	-.017 (.012)	-.012 (.015)
Democrat in house	-.096** (.013)	-.085** (.015)	-.096** (.013)
Partisan in house	.086** (.015)	.078** (.014)	.078** (.014)
Treat × Rep in house			-.006 (.012)
Treat × Dem in house		-.015 (.010)	
Treat × Partisan in house	-.011 (.009)		
Observations	83,740	83,740	83,740
R ²	.071	.071	.071

Notes: Models include demographic covariates and cluster standard errors by household.

[†] $p < .10$, * $p < .05$, ** $p < .01$.

preregistered, this analysis draws on our theoretical understanding that norms surrounding primary participation might shape crossover voting. These views, which Gerber et al. (2017) argue are, at baseline, exclusionary, may be reinforced by housemates registered with a political party who “should” be deciding party nominations. Specifically, we investigate whether the partisan registration status of housemates shapes treatment’s ability to induce crossover voting.

Table 8 reports the household-level heterogeneous effects of treatment on turnout to the Republican primary among undeclared voters who live with at least one partisan registrant, at least one Democrat, and

at least one Republican.²⁹ Table 9 reports the same heterogeneous effects by whether voters have partisan housemates, but estimated for individual-level treatment for voters in multi-voter households.³⁰

Focusing on housemate party registration as a covariate, we find differing associations between partisan housemates and turnout to the Republican primary among voters in control. For household-level models in Table 8, living with a registered partisan

²⁹ About 2% of the sample lives with both a registered Republican and a registered Democrat, but because there are so few of these voters, we focus on the more common cases of living with registrants of one party.

³⁰ These models use IPWs.

TABLE 9 Housemate conditional average treatment effects (CATEs) by individual assignment, 2+ voter households.

	Republican primary turnout	Democratic primary turnout	Overall turnout
Mail + digital treatment	.016 (.010)	.023* (.011)	.022* (.011)
Digital only treatment	.010 (.010)	.018 [†] (.011)	.018 (.011)
Republican in house	-.029 (.039)	-.026 (.030)	-.027 (.030)
Democrat in house	-.055 [†] (.033)	-.024 (.037)	-.055 [†] (.033)
Partisan in house	-.016 (.034)	-.016 (.034)	.004 (.038)
Mail + digital × Rep in house	.007 (.034)		
Mail + digital × Dem in house		-.038 (.026)	
Mail + digital × Partisan in house			-.025 (.023)
Digital only × Rep in house	-.002 (.034)		
Digital only × Dem in house		-.053* (.025)	
Digital only × Partisan in house			-.036 (.022)
Observations	22,506	22,506	22,506
R ²	.073	.074	.074

Notes: Models include demographic covariates and cluster standard errors by household.

[†] $p < .10$, * $p < .05$, ** $p < .01$.

is significantly associated with turning out to the Republican primary at higher rates. Living with a registered Democrat, though, is significantly associated with turning out to the Republican primary at lower rates. For individual-level models in Table 9, living with a Democrat is still consistently associated with lower turnout to the Republican primary, but living with a registered partisan is no longer consistently or significantly associated with turnout.

Importantly, all models in Tables 8 and 9 adjust for the VCI scores and past turnout of voters in our sample, so the association between housemate partisanship and undeclared voters' turnout choices is likely not a product of undeclared voters who live with partisans being more engaged or more Democratic or Republican themselves. Figure 3 shows the distribution of VCI for undeclared voters living with and without (a) Democrats and (b) Republicans. In both cases, far more undeclared voters live without partisan housemates than live with them. And, in both cases, undeclared voters who live with housemates registered with a party do not have VCI scores that skew

lower. Undeclared voters living with Democrats or Republicans are no more likely to support Republicans in general elections than undeclared voters without partisan housemates.

These inconsistent associations between housemates' partisanship and turnout to the Republican primary are the first indications that the social context in which undeclared voters choose to cross over may not operate how we expect. At baseline, undeclared voters may be discouraged or encouraged by partisan housemates to cross over and assert themselves into a traditionally partisan process.

Shifting our focus to treatment effects, we find more consistent, albeit mostly insignificant, heterogeneous effects of treatment among those living with partisans and those not. Whether we estimate treatment effects at the household- or individual level, undeclared voters who live with registered Democrats or Republicans are less effected by treatment and so cross over at slightly lower rates. These heterogeneous effects, though, are rarely significant.

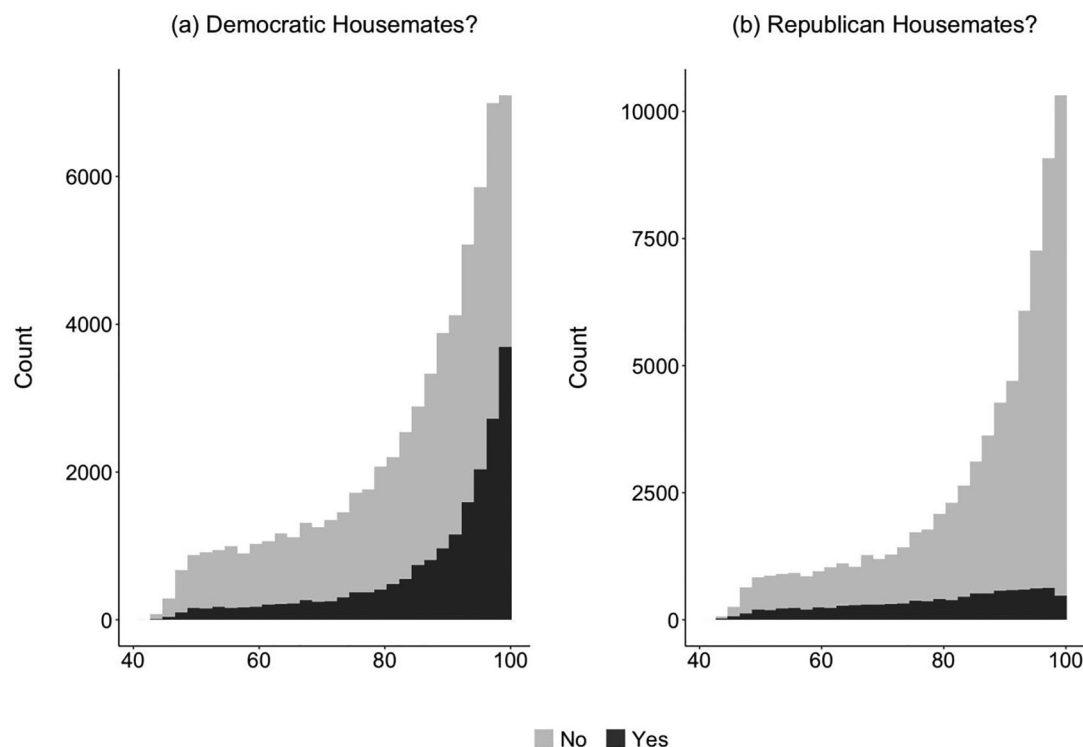


FIGURE 3 Distribution of modeled support for Democrats by whether voters have (a) Democratic housemates or (b) Republican housemates. *Notes:* This figure shows a similar distribution of support for Democrats among voters who live with Democratic or Republican housemates and those who do not.

This is contrary to our hypotheses that Democratic and Republican housemates would affect treatment efficacy differently. Instead, partisan housemates of all types may reduce treatment efficacy. When undeclared voters are not primed to cross over, their housemates registered to a political party do not consistently push them toward or away from crossover voting. But when undeclared voters are explicitly encouraged to crossover vote, their partisan housemates may reinforce exclusionary norms of which voters should vote in primaries, attenuating treatment efficacy.

These findings suggest that our treatment may be effective in part because it changes norms around the acceptability of crossover voting, and that our treatment's ability to change these norms could be moderated by partisan registrants in the household. This would also suggest that barriers to crossover voting may lie in part with partisans who disapprove of the behavior and discourage voters who are eligible to cross over from doing so.

Though this explanation is congruent with our findings, we do not directly test this argument, nor do we measure partisan housemates' attitudes about crossover voting. Another explanation is that undeclared voters who live with partisans are more partisan themselves and so more attached to whichever party's primary they usually turn out to, making treatment

less effective. Still, this would be inconsistent with negative heterogeneous effects on Republican primary turnout among those who live with Republicans and with the distribution of undeclared voters' VCI across types of households shown in Figure 3. The more likely alternative explanation is that these heterogeneous effects are noise within our sample.

Spillover effects

Our contention that treatment effects are in part driven by changing social payoffs of voting is bolstered when we estimate the "spillover" effect of treatment on voters outside our sample.³¹ These registered voters lived in the same household as a voter in our experimental universe but were excluded from that universe because they were either registered as a partisan or were registered undeclared but did not meet the targeting criteria for the experiment.³² All members of our spillover universe could not be directly treated but had a $\frac{3}{4}$ chance of being in a treated household. Partisan registrants in our spillover universe did not have

³¹ As noted, we use the term spillover loosely, in line with Nickerson (2008).

³² For example, they could have had lower levels of modeled Democratic vote choice.

TABLE 10 Turnout outcomes for different registration categories in spillover universe.

	Undeclared voter turnout to Republican primary	Democratic voter turnout to Democratic primary	Republican voter turnout to Republican primary
Treated	.000 (.007)	-.002 (.008)	-.029** (.011)
Observations	20,973	21,005	11,022
R ²	.208	.078	.066

Notes: Models include demographic covariates except for cell likelihood scores which have missingness in our spillover universe. Standard errors are clustered by household.

† $p < .10$, * $p < .05$, ** $p < .01$.

the same turnout choice as our experimental participants but could choose either to vote in their party's primary or abstain. Undeclared spillover voters could turn out to either primary.

Table 10 displays the household-level spillover effects of treatment on primary turnout for different types of housemates. We hypothesized that registered Republicans exposed to treatment would be more likely to turn out to the Republican primary and registered Democrats exposed to treatment would be less likely to turn out to the Democratic primary, because our treatment underlined the relative importance of the two primaries.

Instead, we find that Democratic voters living in households with unaffiliated voters were not differentially likely to turn out to the Democratic primary if their housemates were treated. Similarly, undeclared voters excluded from our experimental sample but living in experimental households were not differentially likely to turn out to the Republican primary when their housemates were treated.

However, we find that Republican voters living in experimental households were significantly less likely to turn out to the Republican primary, by 2.9 percentage points, if their undeclared housemates were treated. Given the usual magnitude of spillover effects, this constitutes a substantial reduction in turnout. Instead of being spurred into more participation when their housemates are told that the Republican primary is important, Republican housemates disengage from their party's primary. This result corroborates our intuition that shifting beliefs about pivotality is not the only causal mechanism. After all, a pivotality-driven effect would suggest registered voters would shift uniformly away from the Democratic and toward the Republican primary.

The negative effect on Republicans is striking, and we offer three potential explanations. First, the treatment could serve to demobilize by de-legitimizing the primary in some way, suggesting that Democratic or independent voters are improperly participating or that the primary is rigged. This mechanism could further operate through social norms around primary voting. Second, the treatment could serve to prime political conflict in politically mixed

households by highlighting the Democratic partisan identity of one member of the household. The negative, though imprecisely estimated, interaction effect between treatment and Republican housemates reinforces this possibility. Third and finally, the treatment could have signaled negative information about either major Republican candidate, reducing turnout through cross-pressure among registered Republicans (Mutz, 2002). For Trump supporters, the treatment presents straightforward negative information about him. However, given that informing voters that a co-partisan candidate is antidemocratic does little to change vote choice (Graham & Svobik, 2020; Walk et al., 2024), we believe treatment is unlikely to demobilize those same voters. More plausibly, treatment could have served as a powerful and clear partisan group cue that Nikki Haley was supported by Democrats (Broockman, Kaufman, & Lenz, 2024; Lupia, 1994). Republican voters who supported Nikki Haley could have been cross-pressured by this information, reducing turnout.

Post-election survey

The post-election survey confirmed the partisanship and vote choices of undeclared voters in our experimental universe in both treatment and control who voted in the Republican primary. As designed, respondents were overwhelmingly independents who leaned toward the Democratic Party; 34% of respondents identified as true independents, 35% as independents who lean towards the Democratic party, 4% as weak Democrats, and 10% as strong Democrats. Also as designed, respondents were likely to support Democrats in general elections; 74% of respondents reported voting for Kamala Harris in the 2024 general election.

What's more, 78% of these crossover voters supported a candidate other than Trump in the Republican primary and, among Kamala Harris supporters in the general election, 95% voted against Trump. Among those who voted against Trump in the primary, 62% said the best reason for their vote choice was, "The candidate would prevent another candidate I disliked

from winning the primary.” Only 1.7% said they supported a Republican primary candidate because that “candidate would give the Democratic Party the best chance of winning the general election.”

This evidence reinforces two assumptions. First, individuals who appeared to be crossover voters in the voter file—that is registered undeclared, modeled Democratic general election voters who voted in the Republican primary—are actually crossover voters. Second, when voters crossed over, they overwhelmingly hedged by voting for the more moderate candidate rather than raided to give their party a better chance in the general election. These results parallel our pre-election surveys suggesting that Democratic general election voters chose Haley over Trump by overwhelming margins.

DISCUSSION

We provide, to our knowledge, the first field-experimental evidence of successfully prompting voters likely to support and identify with the candidate of one party in the general election to turn out to another party’s primary election. A modified version of a standard GOTV treatment is capable of increasing crossover voting and support for the more moderate opposing party primary candidate, among a sample of registered undeclared, but Democratic-leaning and Kamala Harris-supporting, voters in the 2024 New Hampshire Republican presidential primary.

Our results show that it is possible to encourage crossover voting in the real world, even in the type of crowded information environment that often leads to reduced treatment effects (Green & Gerber, 2019; Green, Leavitt, & Markovits, 2024). As more voters become concerned about the state of American democracy but are unwilling to discipline their own party’s democratic transgressions, our results show that partisans are willing to forgo their say in their own party’s primary in order to increase the possibility that the opposing party nominates a more moderate candidate.

We note that there are neither theoretical nor empirical reasons why this style of intervention cannot extend across party lines. Like Democratic partisans, Republicans express concern at the extremism of the other party (both in terms of policy and lack of democratic commitments) and in future competitive Democratic primaries, they will have opportunities to engage in the same crossover behavior we studied among Democrats. However, there are several factors that make the context of the New Hampshire primary unique. The intensity of media coverage and campaign spending surrounding the primary may have blunted our treatment effects, because control group voters were subject to abundant messaging surrounding the

importance of the Republican primary. However, that same intensity of coverage simplified several informational challenges to crossover voting by informing undeclared voters they were eligible to vote in either primary, obviating difficulties faced in lower salience primaries of knowing which candidate was more moderate, and alleviating coordination challenges. Finally, the lack of a competitive Democratic presidential primary reduced the opportunity cost inherent to crossover voting.

We also use these results, along with pre- and post-election surveys, to refine our theoretical understanding about what drives and what inhibits crossover voting. Crossover voting involves two choices: Who to support in the opposing party’s primary, and which primary to vote in (or to abstain). In the abstract, surveys reveal that would-be crossover voters have strong normative concerns about, and therefore little appetite for, “raiding,” or voting for a less preferred candidate in the opposing primary to bolster your own party’s chances in the general. In surveying voters in our sample who did crossover during the New Hampshire primary, we show these voters instead “hedge,” or support the more moderate candidate to protect against extremism.

Because this was an experiment to induce a unique behavior, treatment was intentionally bundled, providing information about electoral institutions, candidates, and social pressure. Though we do not test each component of the treatment separately, we believe two factors, the relative competitiveness of partisan primaries and social norms about who should vote in primaries, are part of what influences voters’ decisions to cross over. First, treatments not only increased undeclared voters’ turnout to the Republican primary, but decreased their turnout to the Democratic primary, suggesting that around a third of crossover voters would have turned out to the Democratic primary if untreated. Already prepared to vote, undeclared voters switched to which primary they turned out as treatment emphasized the importance of the Republican primary. Second, though not significant, we find a consistent negative relationship between treatment efficacy and living in a household with a registered partisan. We posit that treatment may induce less crossover voting among undeclared voters living with registered partisans because those partisans enforce norms about who should have a say in partisan primaries. Third, Republicans living with treated undeclared voters are significantly and substantively less likely to vote. Despite being exposed to information about how important their primary is, Republicans turn out at lower rates, perhaps because they are put off by non-Republicans voting in their primary or non-Republicans supporting Republican candidates.

Contrary to our expectations, undeclared voters’ own modeled partisanship does not influence their

responsiveness to encouragements to crossover vote. Using linear and nonlinear estimation strategies, we find no heterogeneous effects of treatment based on voters' likelihood to support a Democrat in the general election or by their past turnout to Democratic presidential primaries. Treatment encouraging crossover voting is not inhibited as voters are more reliable Democratic general election voters. Additionally, education does not influence treatment efficacy, with more educated voters no more affected by treatment. We add to past work that yields conflicting results as to whether educated voters are more likely to act strategically (Duch & Palmer, 2002; Eggers & Vivyan, 2020; Miranda & Meloni, 2016). However, one caveat is that measurement error inherent to our modeled covariates could depress heterogeneity because some individuals modeled as Democrats or as educated in fact are not.

That encouragements to crossover vote are effective in changing voting behavior has important implications for campaign spending, electoral reform, and electoral competition. First, even as party campaign committees pour more money into ensuring the opposing party nominates a weak candidate, voters have demonstrated a strong dislike of these machinations (Hussein, Lee, & Wheeler, 2025) and little willingness to participate in these efforts, instead promoting the opposing party's moderate candidate. Second, increasingly popular open primary reforms may have failed to deliver the more moderate nominees they promised (Ahler, Citrin, & Lenz, 2016; Bullock & Clinton, 2011; Geras & Crespin, 2018; McGhee et al., 2014), because crossover voting is uncommon at baseline. With more widespread encouragements to crossover vote, the composition of primary electorates can shift, and with it, the incentives for candidates to moderate. Finally, voters who are effectively disenfranchised because they belong to the minority party in a one-party district or state can use crossover voting to make their voice heard in the decisive primary election.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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